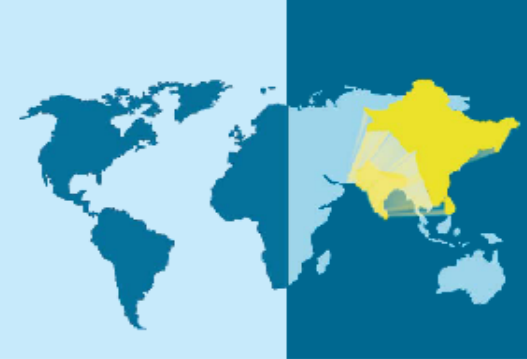




IPA CONVENTION 2007

Credible India: Sustaining Competitive Advantage



Information Technology
to achieve Operational
Excellence

Rudiger Dorn,

Director, Worldwide Life Sciences Group, Microsoft

The role of IT to achieve Operational Excellence

Operational Excellence in Pharma Manufacturing

Rüdiger Dorn

Managing Director Pharmaceuticals

WW Manufacturing

Microsoft Corporation

What is Operations Excellence?



“We’ve found the key to productivity. It’s Fred, down in the shop. He makes the stuff.”

**Integrating processes, information and people
in a compliant environment**

People Drive Business Results



**Develop
Customer
Relationships**

**Drive
Innovation**

People

**Improve
Operations**

**Build Partner
Connections**



Pharma focus returns to growth



1

Top line growth



2

Cost cutting

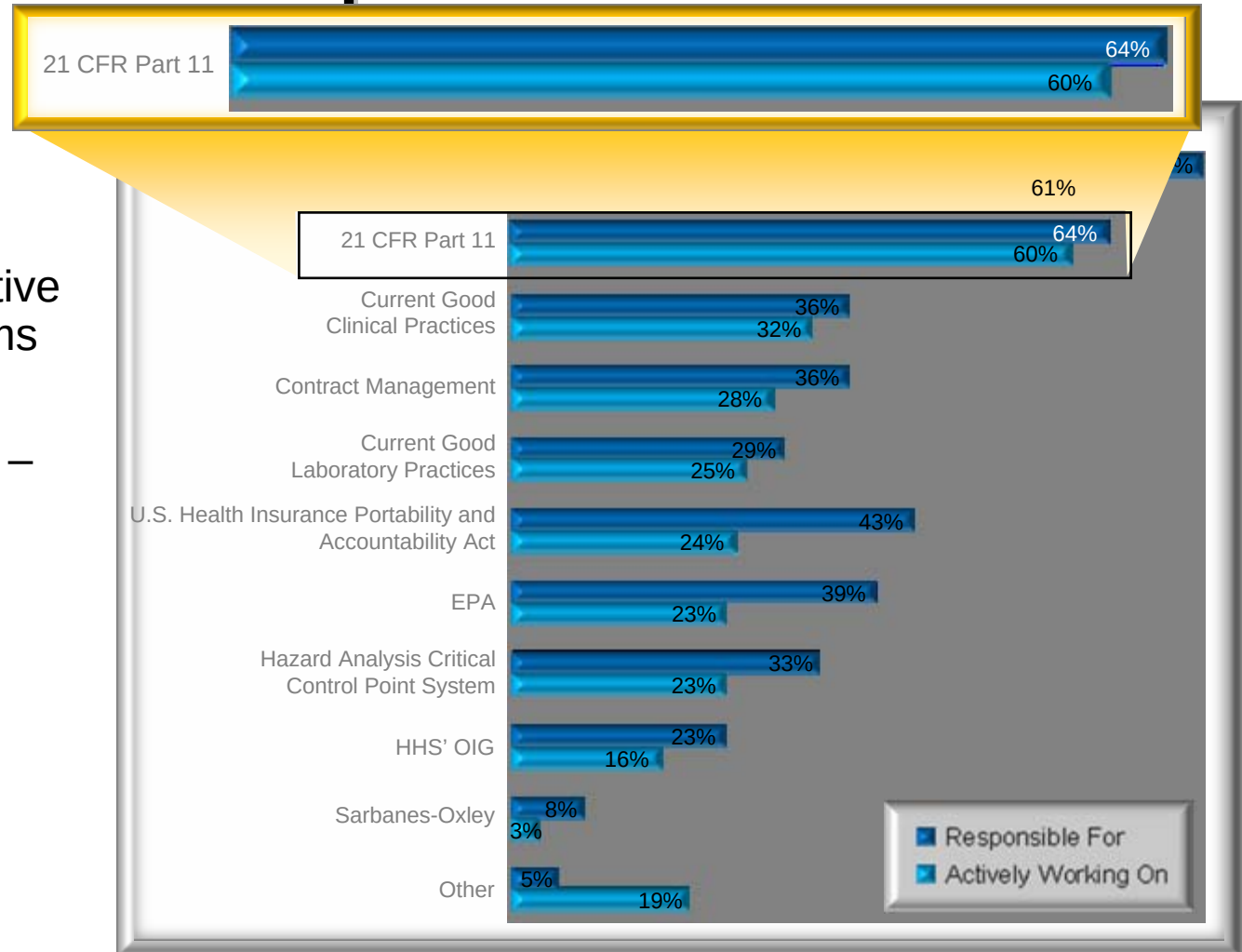


3

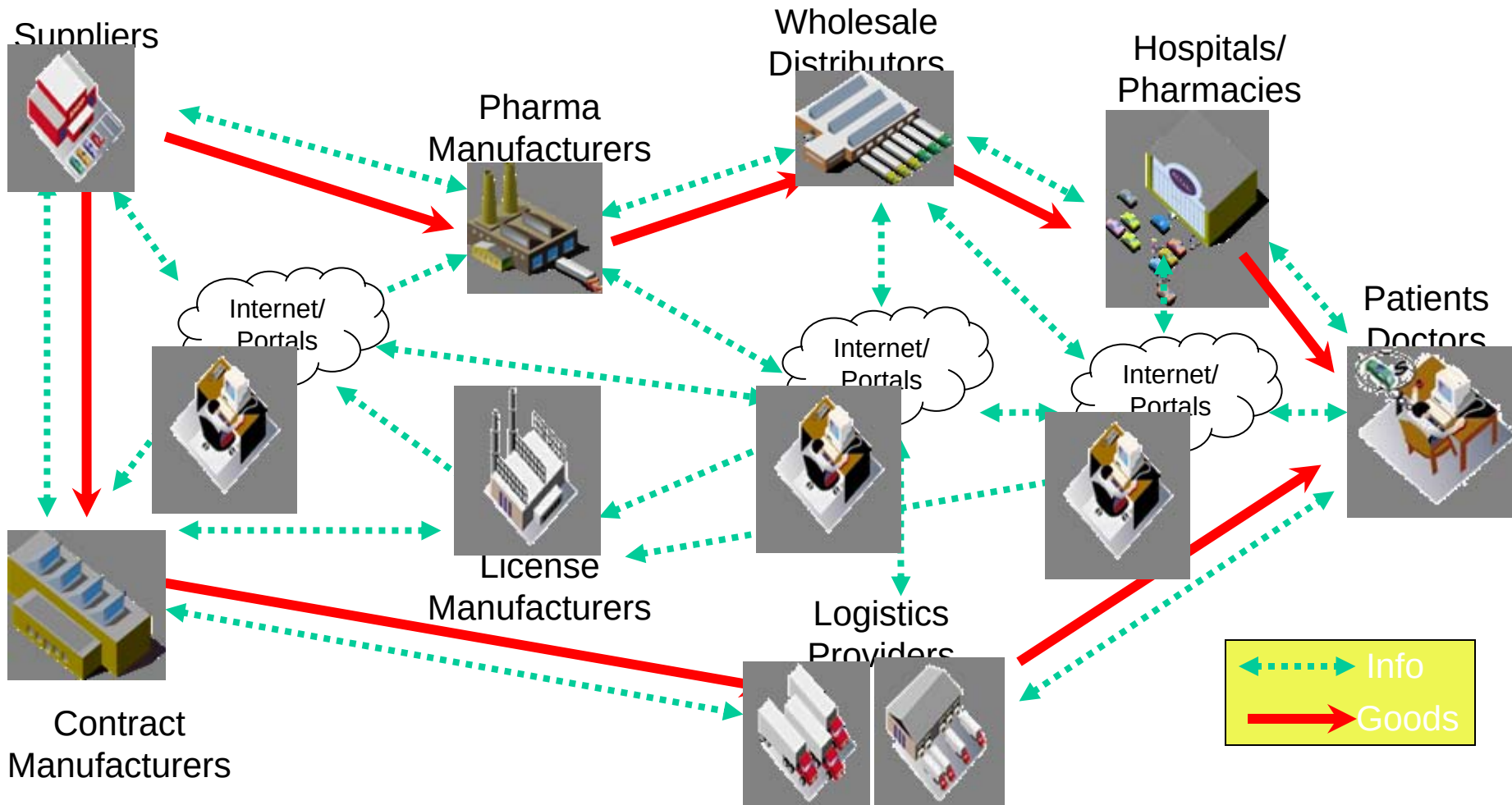
Compliance with regulations

The focus is on industry specific compliance

- 21 CFR Part 11 is the second most reported compliance initiative in life science firms
- The number one initiative – cGMP – is also subject to 21 CFR Part 11 regulations when electronic documentation is used



The Supply Chain is Evolving to a Collaborative Supply Network

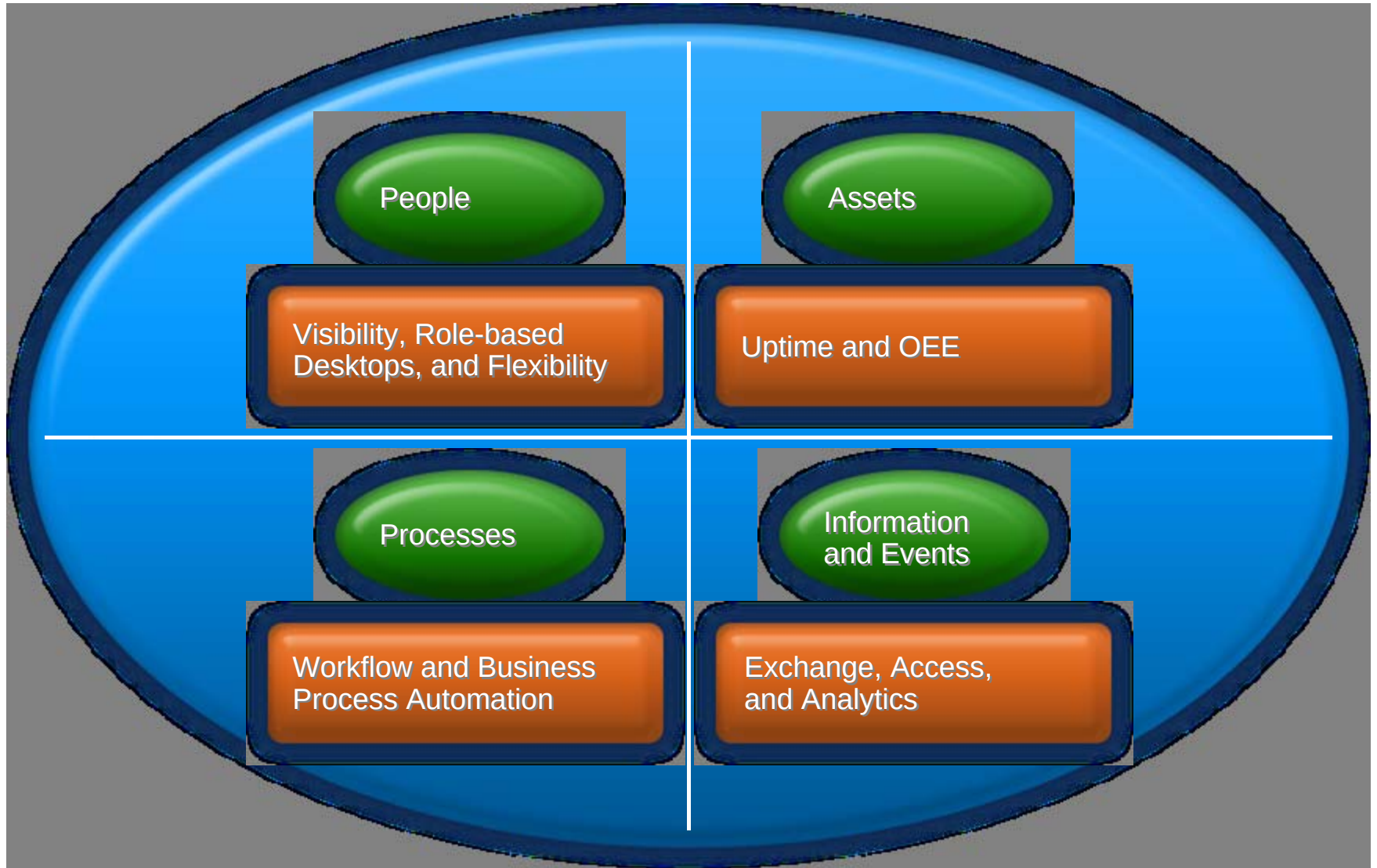


Plant Operations Challenges

- Reduced cycle times to bring products to the market (new facilities)
- Flexibility and agility to build and move capacity
- Visibility: “You cannot improve what you cannot see” – Dr. Deming
- Common key performance indicators across plants
- Improved yield/throughput
- Inventory management and WIP reduction
- Regulatory compliance for first validation AND subsequent re-validations
- Consistent product quality – prevent problems before they happen
- Sharing and retaining knowledge between plants and systems for increased flexibility and agility
- Simplification through standardization



Operational Performance Initiatives to date



The promise of information transparency

Improving organizations by providing business insights to **all** employees, leading to better, faster, more relevant decisions

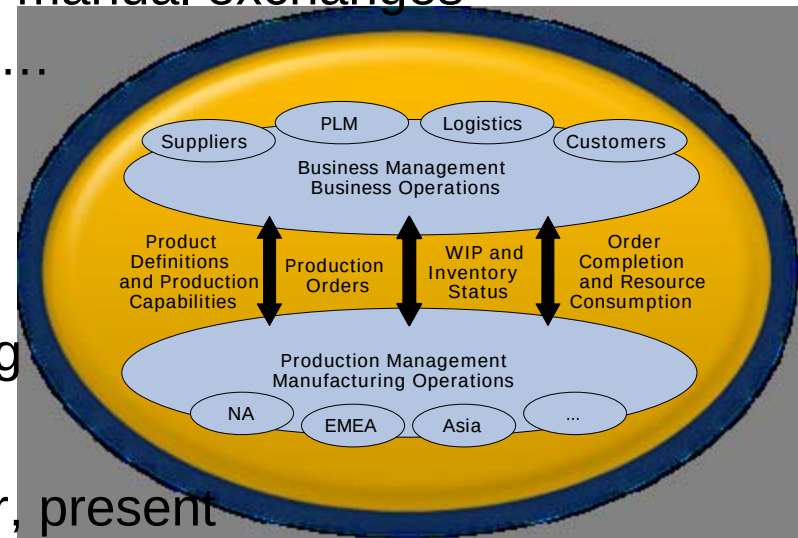
- Add value to every decision
- Increase the business performance of organizations
- Bring BI into collaboration & business process



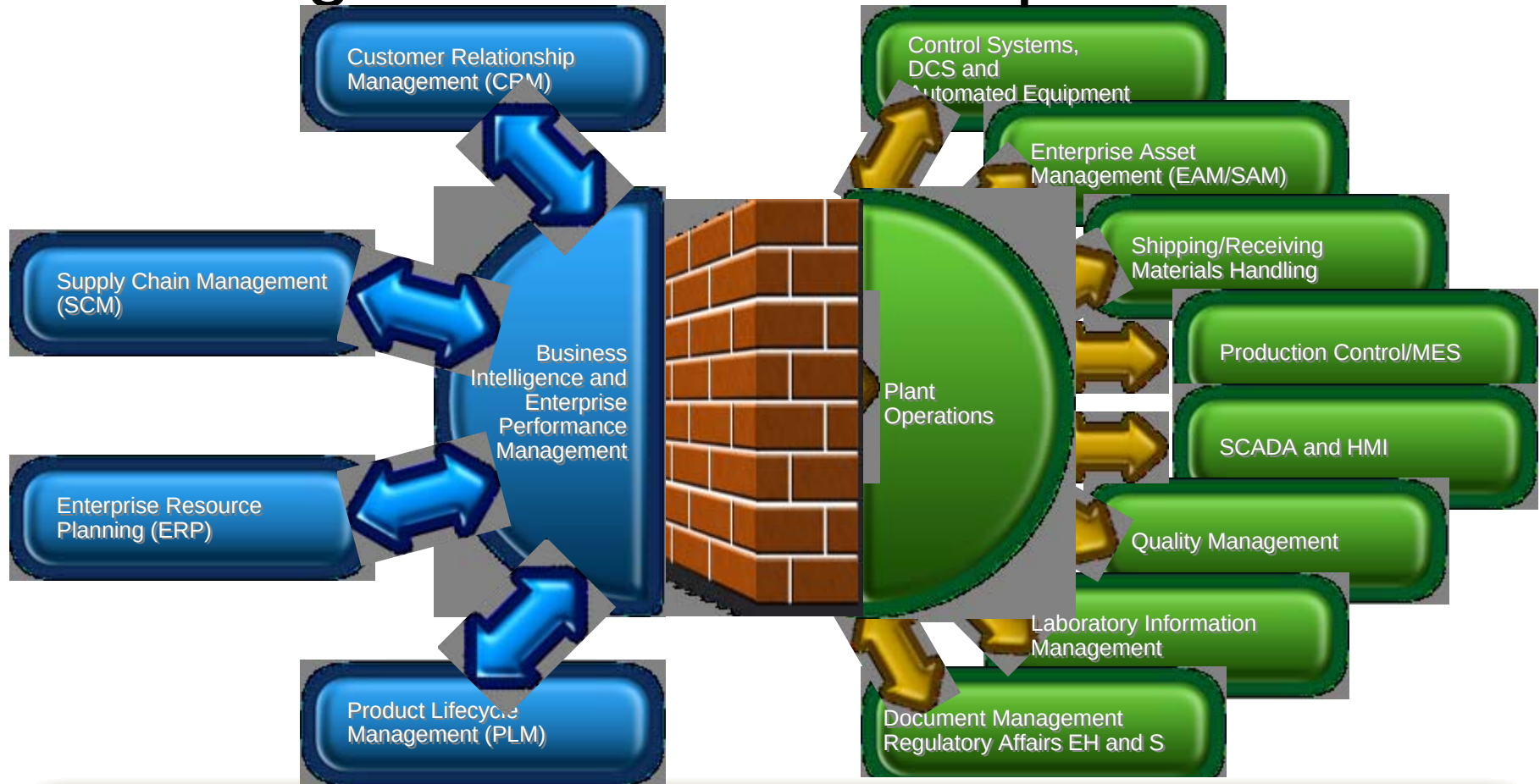
Right Information

The challenge to find the right data

- Issues
 - Rapidly growing data stores
 - Locked within applications, domains, manual exchanges
 - Confidentiality, compliance, security,...
- Improve
 - Management
 - Access, visibility, and utilization
 - Create knowledge and understanding
- Strategies
 - Integrate, aggregate, federate,...filter, present
 - Application-specific analytics
 - Analytic tools

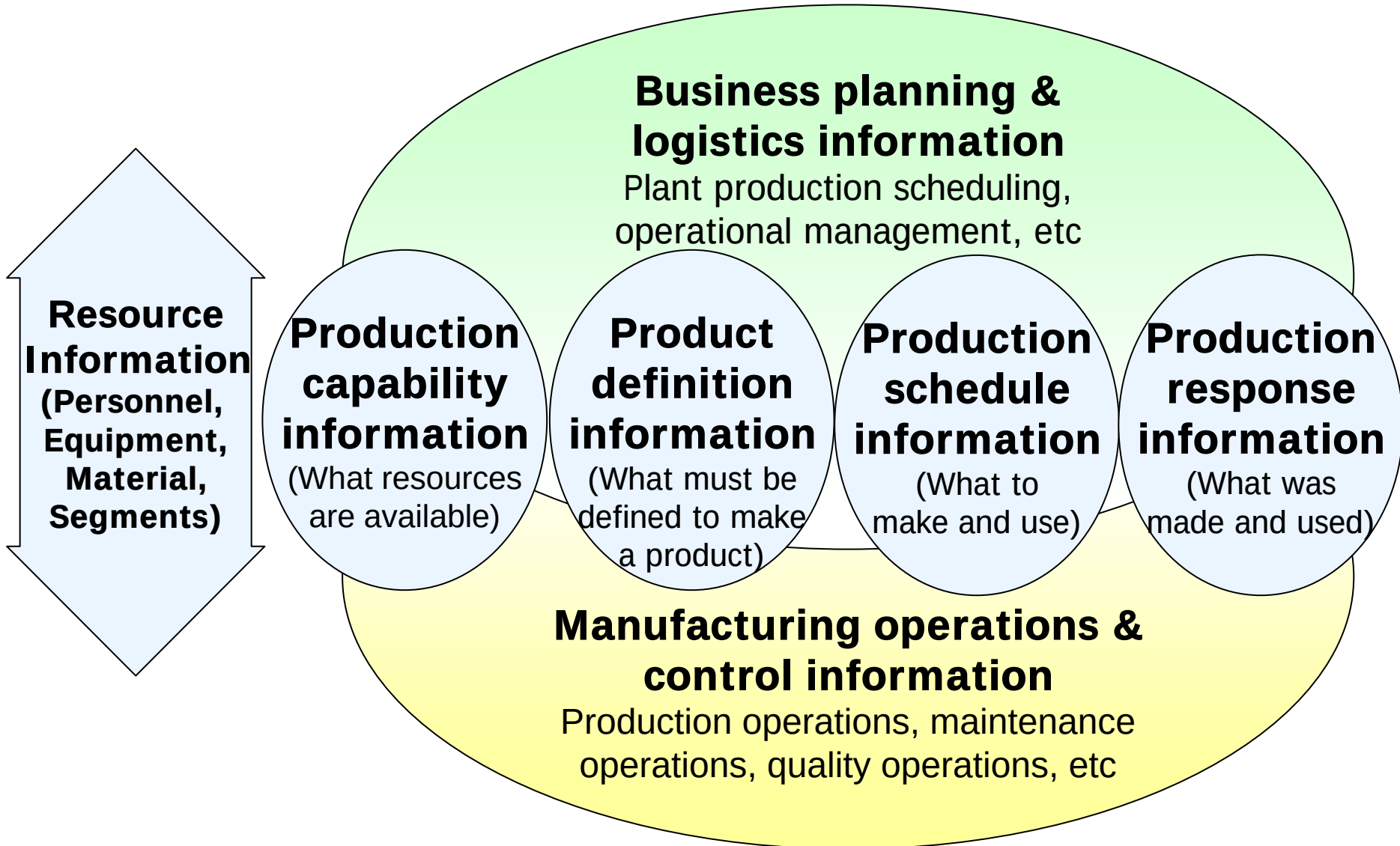


Enterprise Applications Integrated With Plant Operations



Missing link: Plant operations impact on results inventory, processes, orders, resources, status, downtime, products, lab results, exceptions

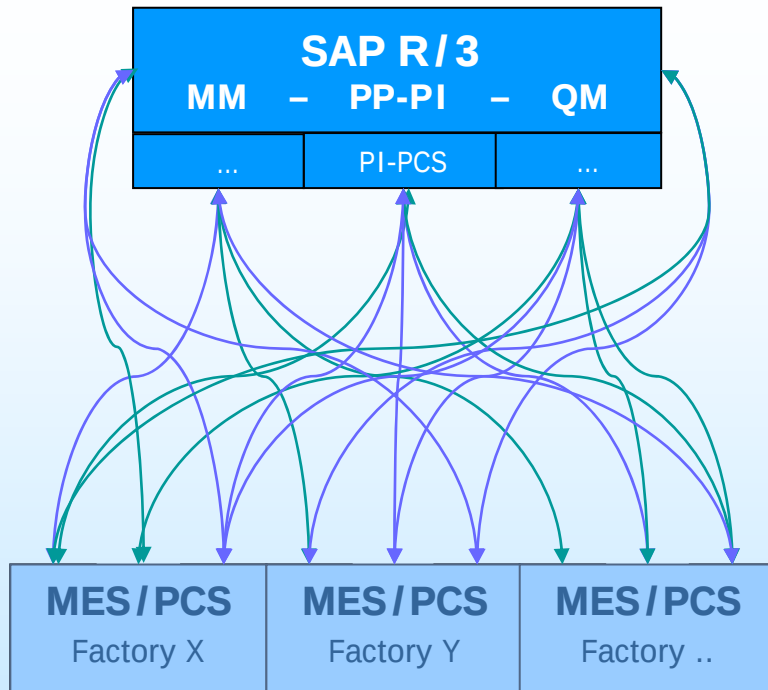
ISA-95 Data Exchange Standards



Reducing costs in Operations

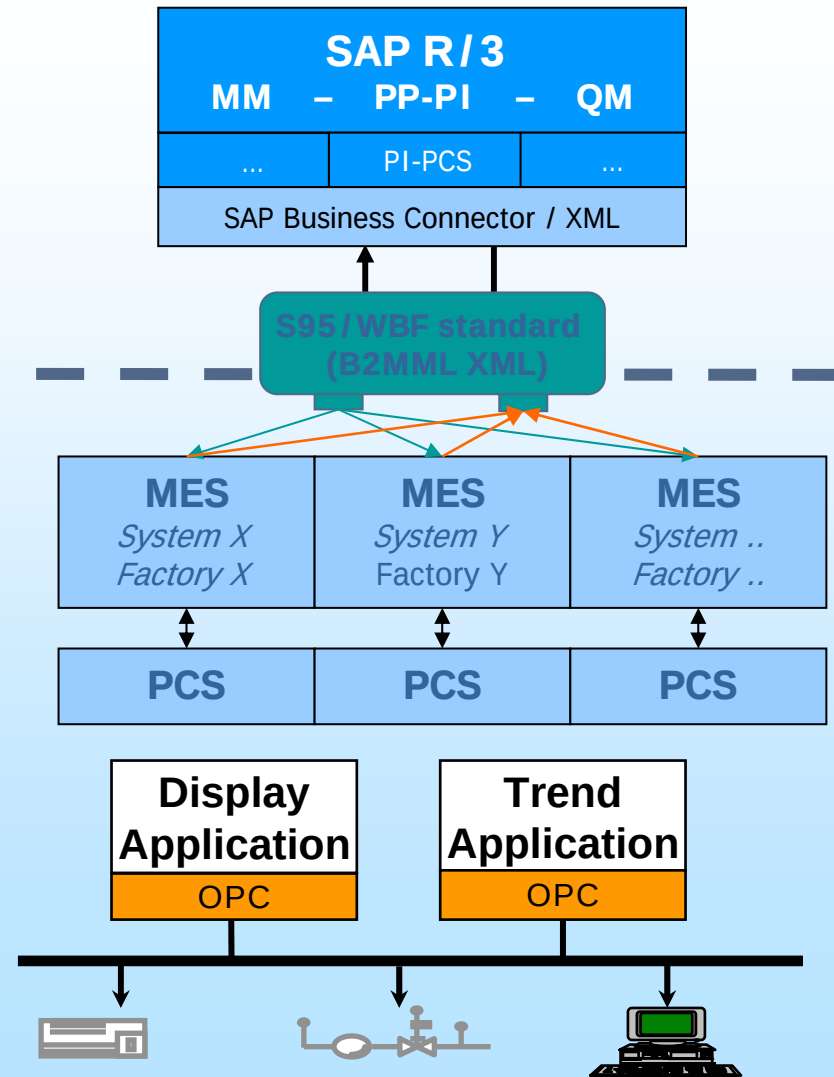
Scenario 1 – to be avoided..

Using several vendors proprietary "standard" solutions

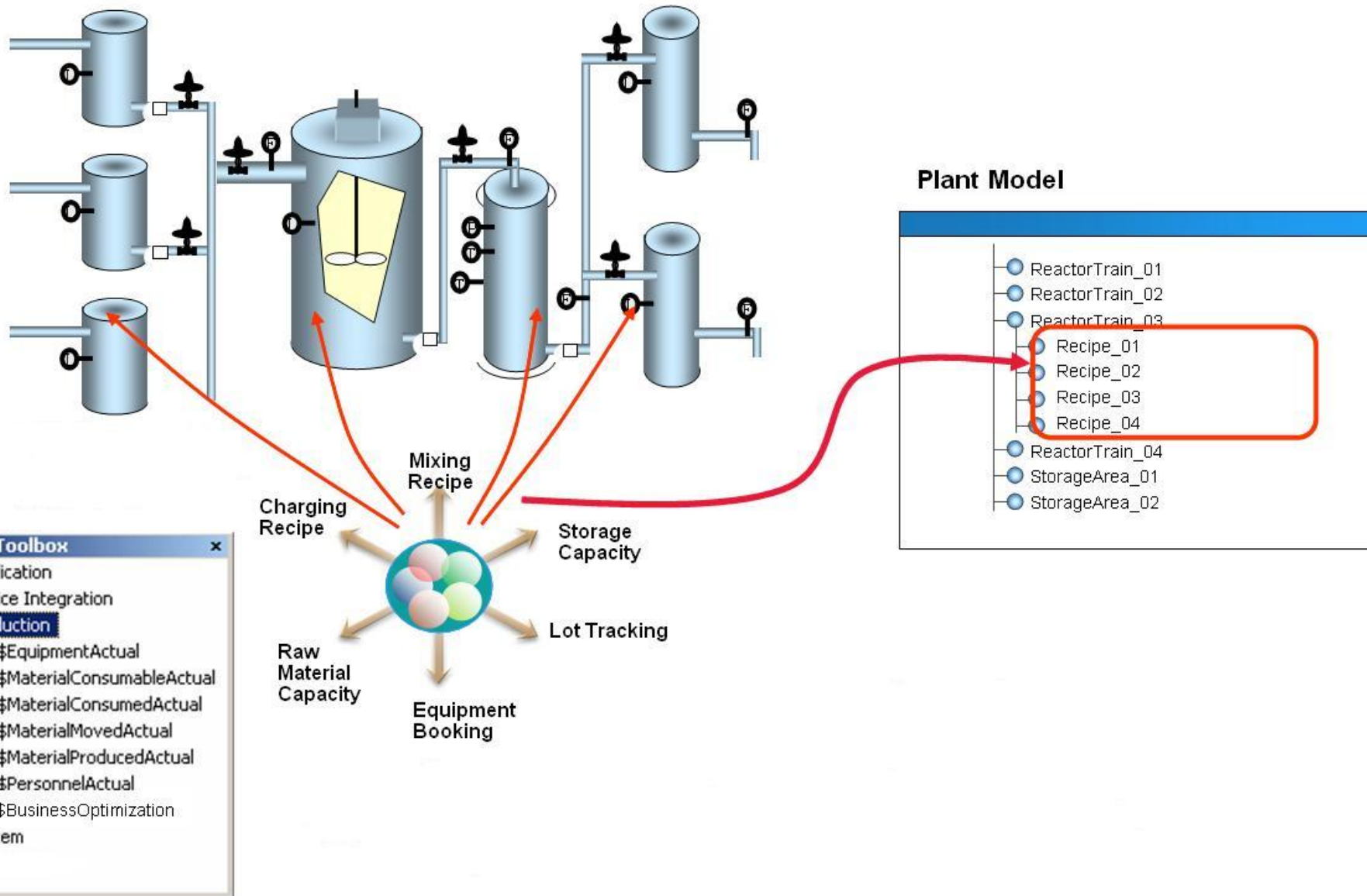


Scenario 2 – to be achieved:

Using one generic standard solution



Modularization of information

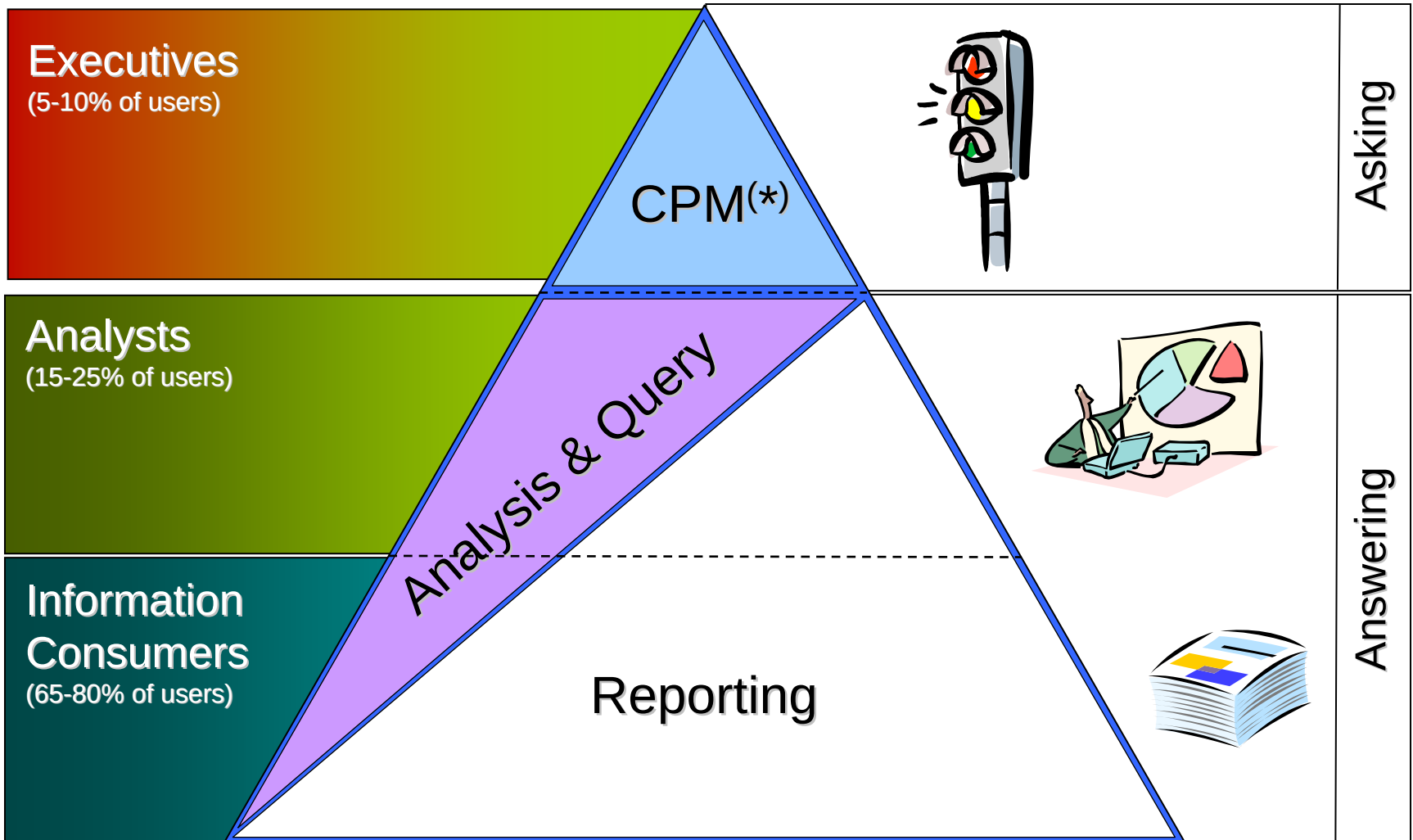


Some Users of the ISA-95 Standard

- Energy
 - ExxonMobil, Inc
 - Kuwait National Petroleum Company
 - ExxonMobil Neftegas Limited
 - BP
 - British Gas
 - Caspian Pipeline Consortium
 - U.S. Department of Energy
 - GasOil Resources LLC
 - Sakhalin Energy
 - Westinghouse Savannah River
- Discrete Manufacturing
 - Advanced Micro Devices
 - JDS Uniphase
 - Computer Sciences Corporation
 - GE Lighting
 - GE Aircraft Engines
 - Lucent
 - Alcatel
 - Cummins
- Other
 - International Paper
- Life Sciences
 - Eli Lilly and Co.
 - Merck
 - Genentech
 - Roche Diagnostics
 - TEVA
- Food and Beverage
 - Nestle
 - Coca-Cola North America
 - Arla Foods
 - United Biscuits Verkade Zaandam
 - Van Dijk Food Products Zeewolde
 - Cervceria Polar
 - GumLink
 - SAB Miller
 - Frito-Lay
 - Red Gold Tomatoes
- Consumer Goods
 - P&G
 - British American Tobacco
- Chemical
 - DOW Chemical
 - Dupont
 - SABIC (Saudi Arabia)
 - Polaroid

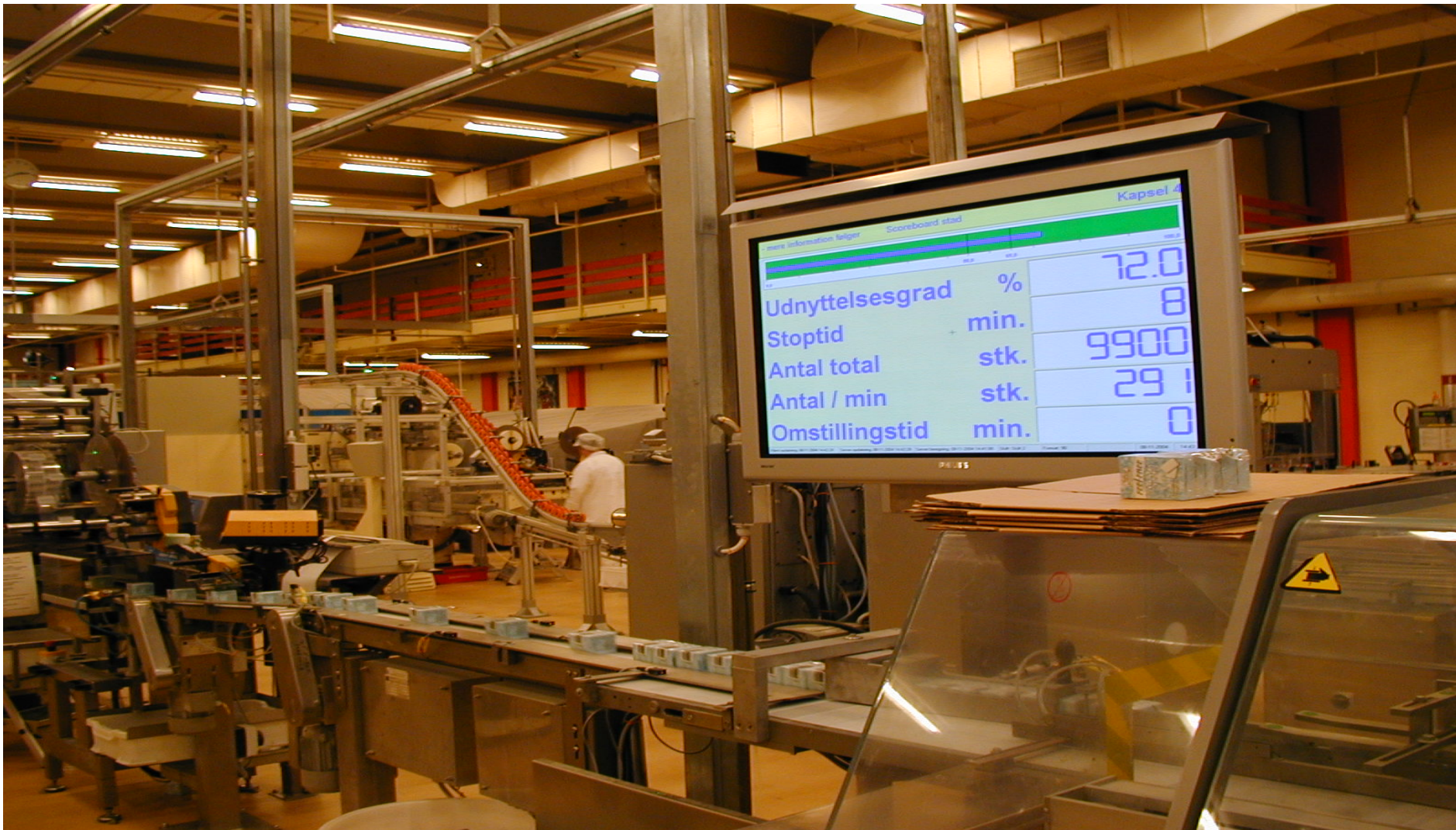
Right time

Information needs differ

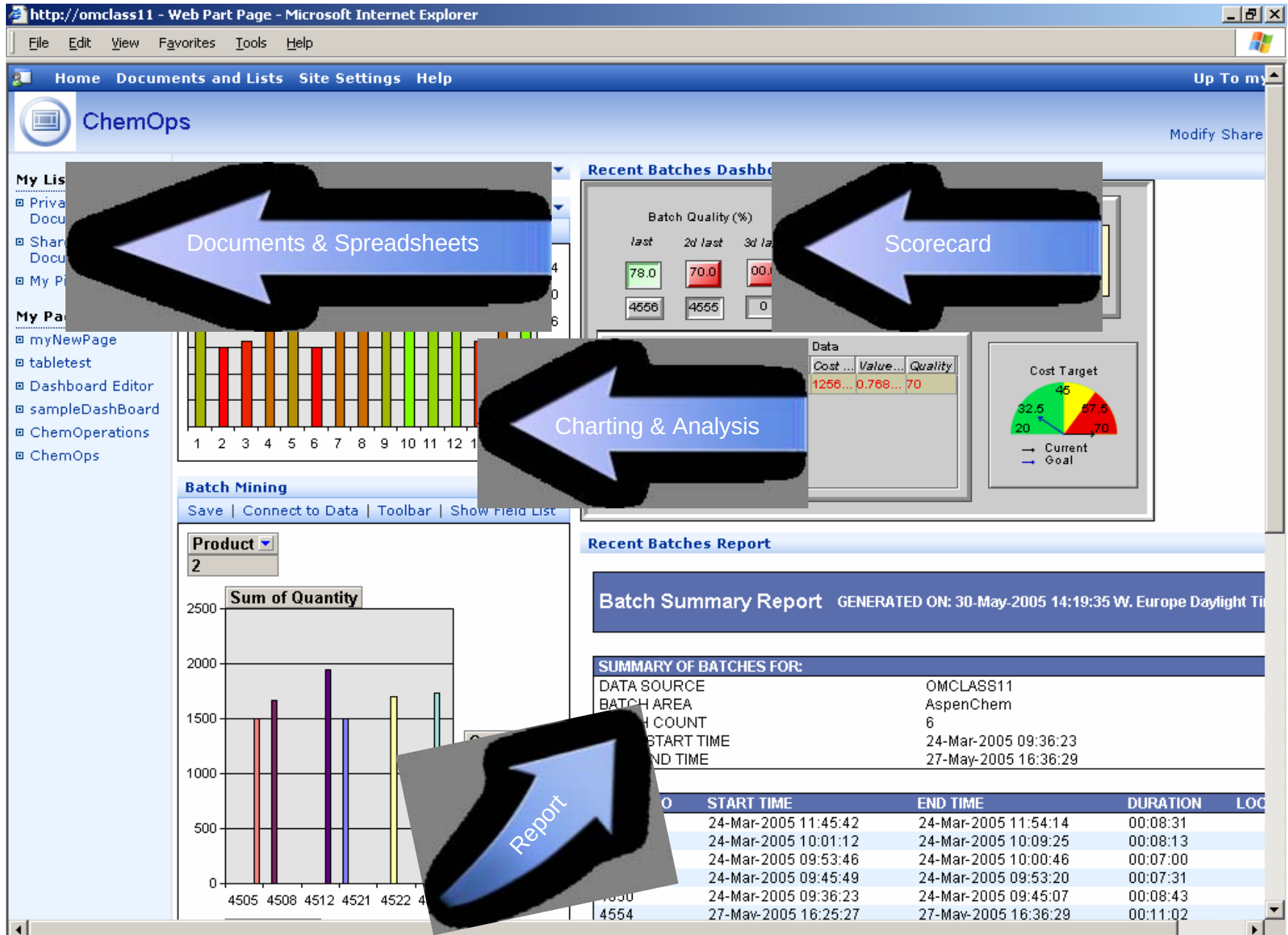


(*) CPM = Corporate Performance Management

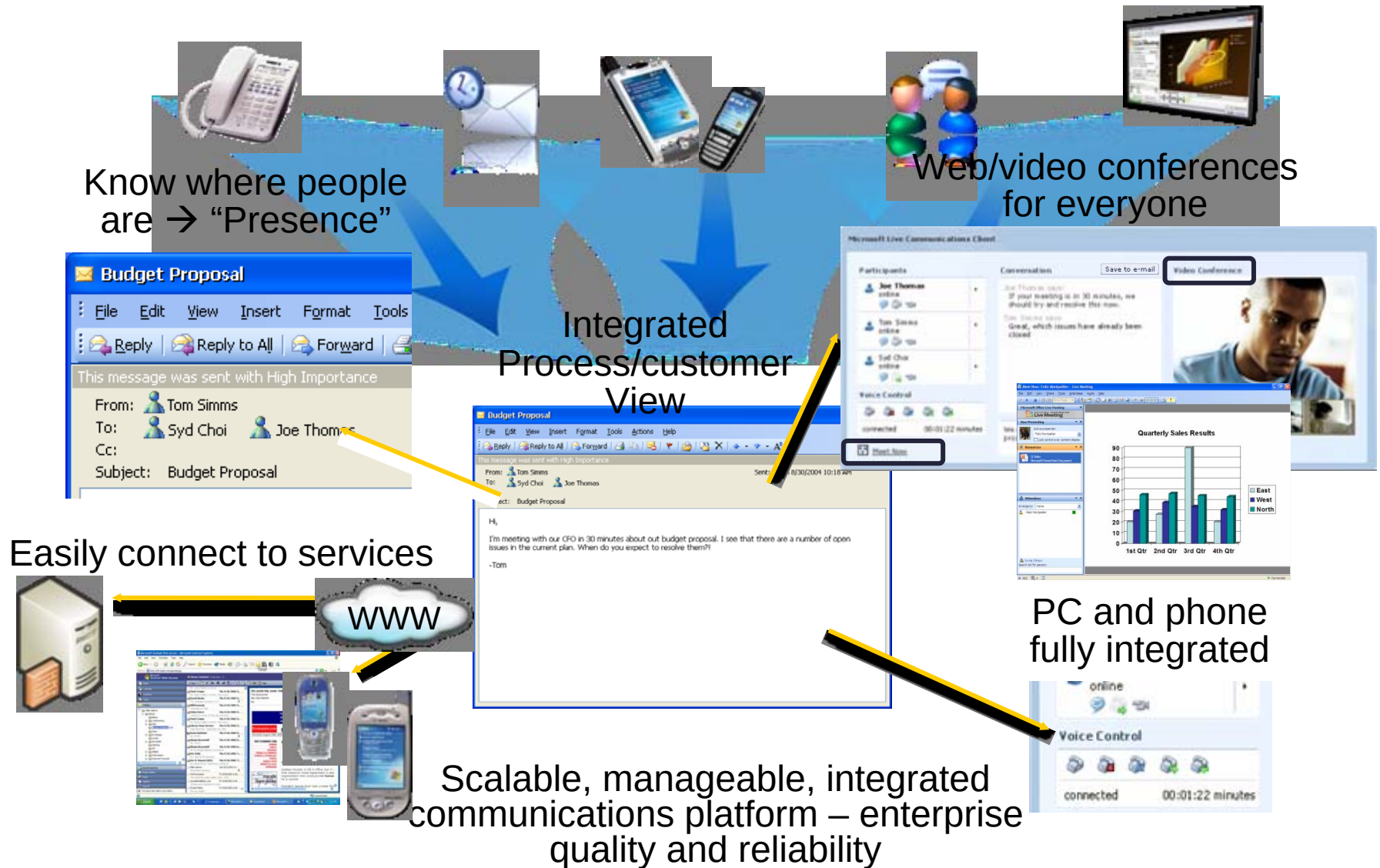
Plant floor intelligence at the production line



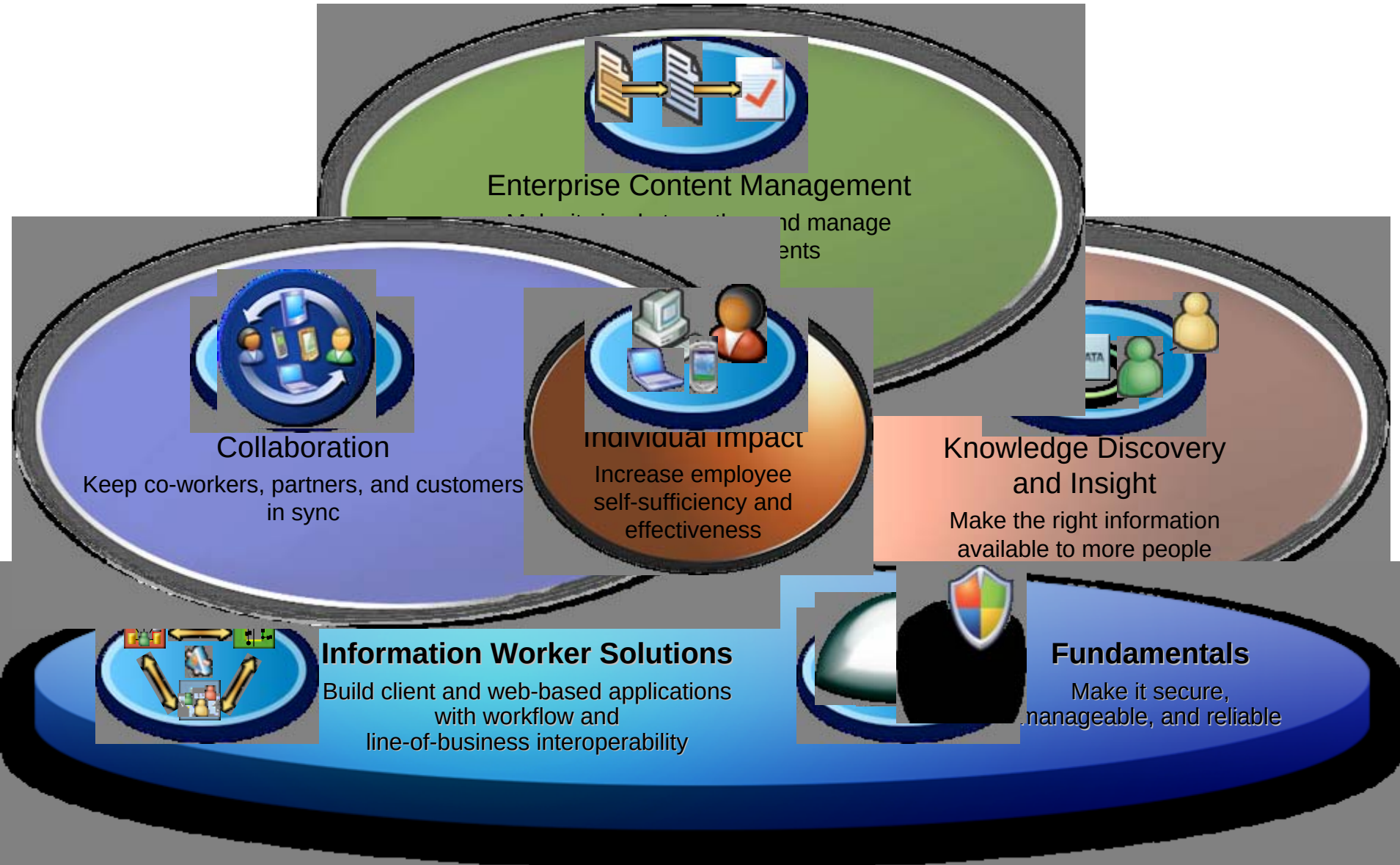
Right Format: The Need for contextual insight



Integration of Information with Collaboration



Comprehensive Collaboration Strategy

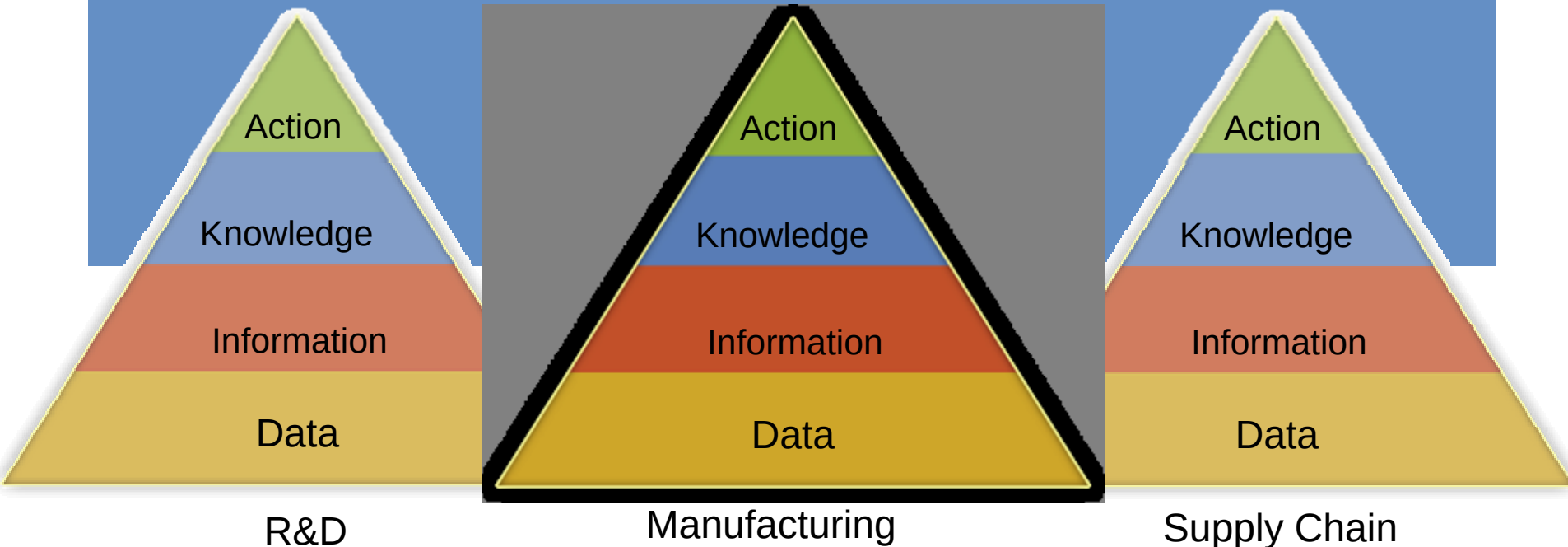


The Productivity Vision

Make the knowledge accessible – real time!



Collaboration layer to exchange knowledge and access data if necessary



Changing Relationship Between the Business and IT

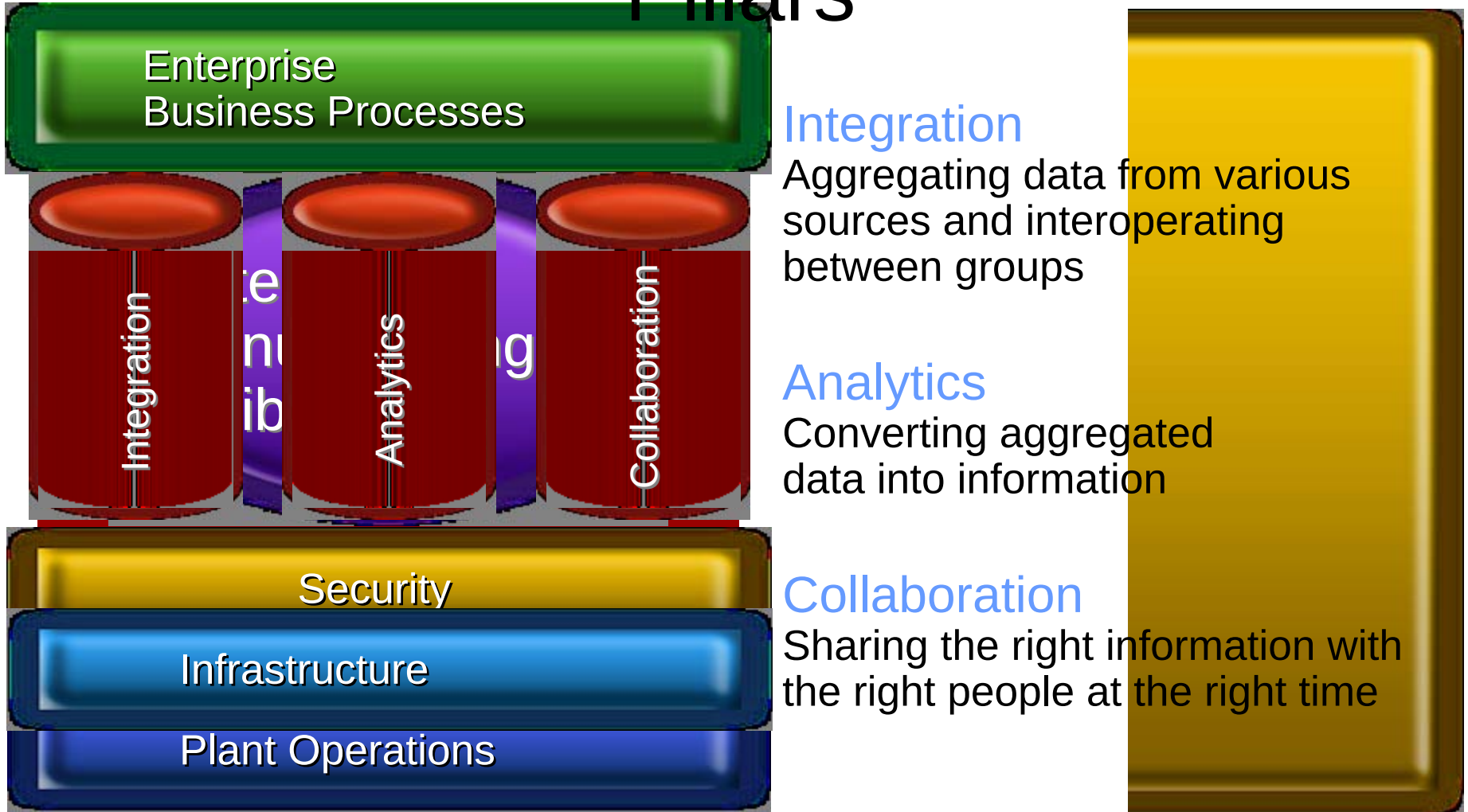


- Business users manage the application
 - Rules – e.g. allocations, business relationships, alerts
 - Workflows – e.g. planning and forecasting cycles
 - Content – e.g. accounts, business views, roll-ups
- IT centrally manages security and information
- Increased business flexibility, declarative model-based applications, and reduced development time

Microsoft Manufacturing Vision

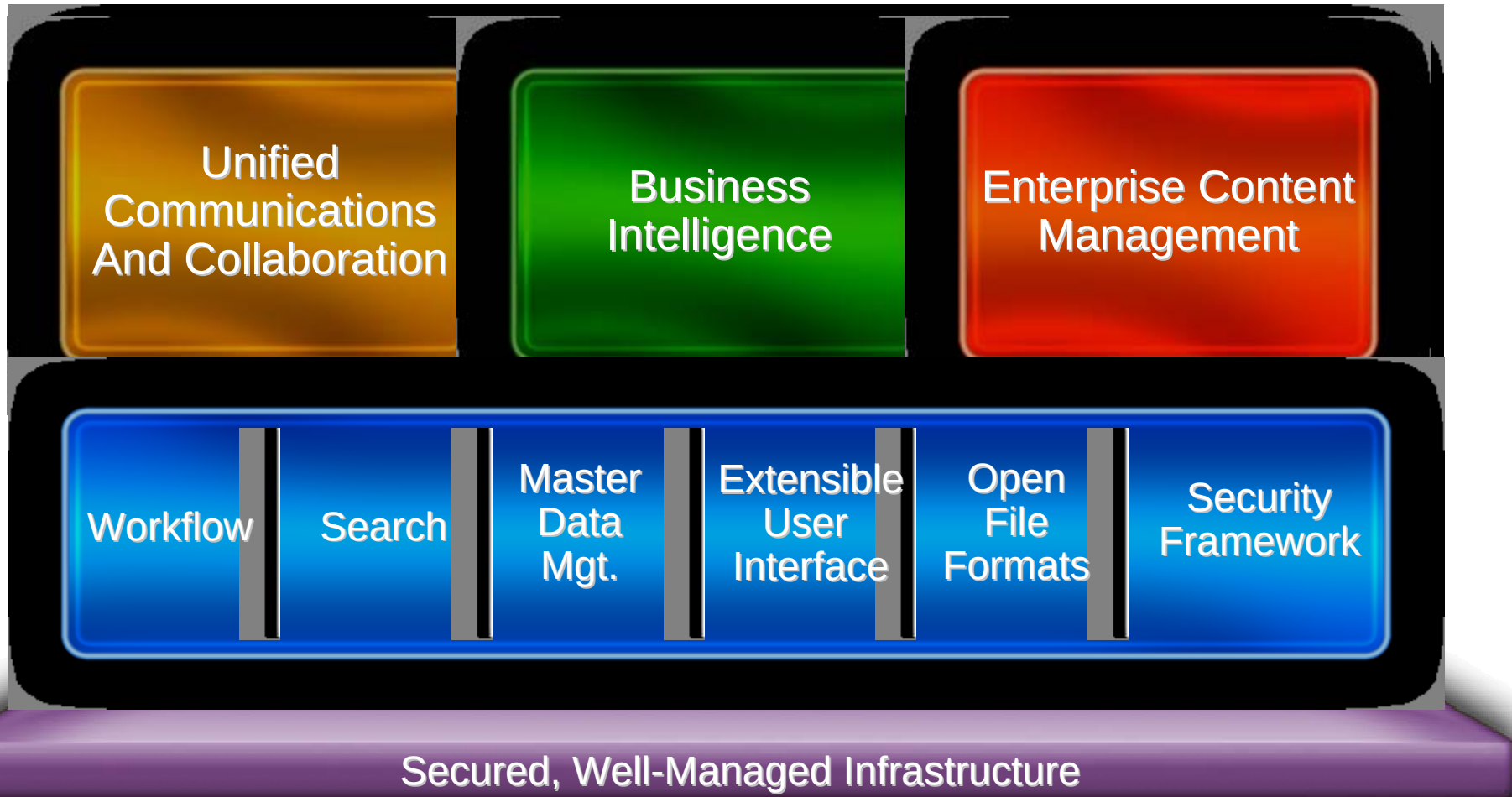
To accelerate our customers' insight into their businesses and empower their innovation by delivering Microsoft and Microsoft partner solutions that will enable their people to improve their ability to achieve peak performance

Microsoft Plant Floor Visibility Pillars

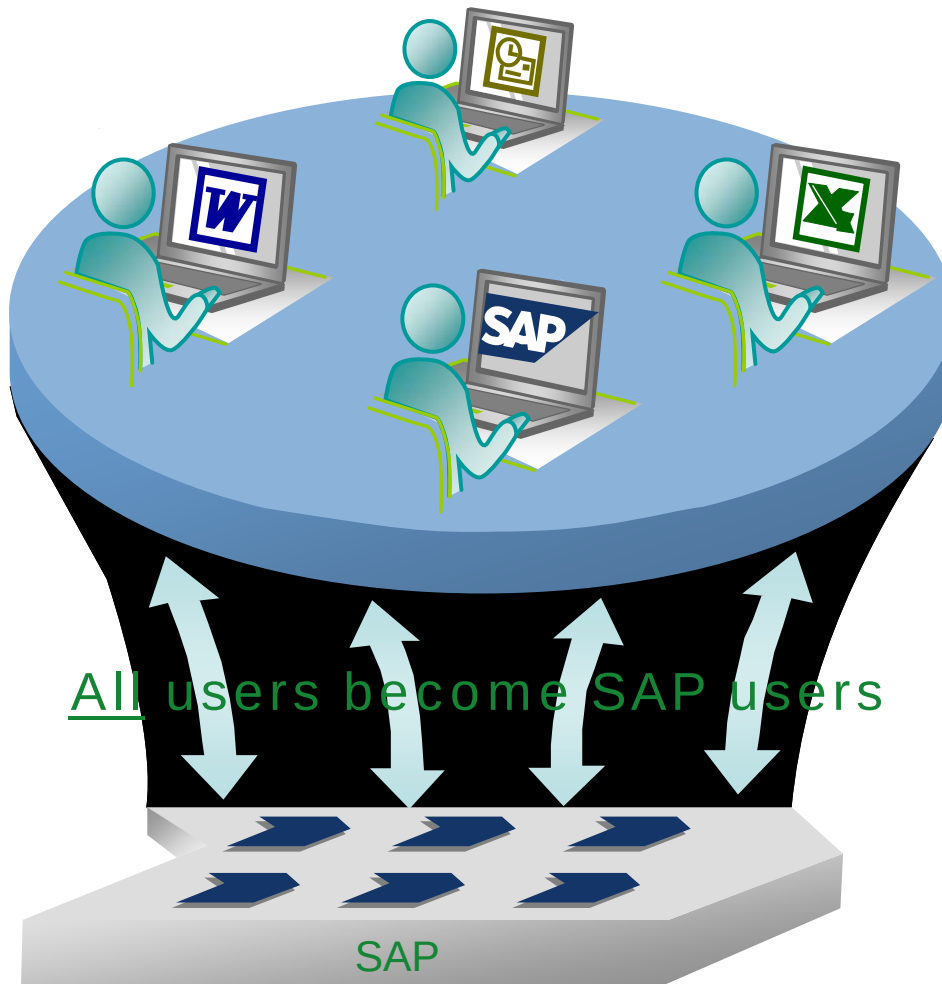


Infrastructure for Operational Excellence

Streamline the way people do business



Microsoft and SAP for Operational Excellence



- Connects to critical information via familiar user interface
- Allows to combine ERP and Shopfloor data
- Unites structured and unstructured data and processes
- Choice of access and 24/7 availability
- Simplifies business process access for greater agility

